



The new SOLPS-ITER code package



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ABSTRACT

The SOLPS-ITER package represents a renewed coupling of the up-to-date parallelized EIRENE Monte-Carlo code and the B2.5 edge plasma solver. Each part of the fluid-kinetic code system has been dismantled and analysed. For SOLPS-ITER, full backwards compatibility with previous SOLPS4.3 ITER reference simulations is critical. We present the results of the ongoing comprehensive benchmark activity and discuss how SOLPS-ITER can be tuned to reproduce ITER simulations performed with SOLPS4.3. In particular in SOLPS-ITER a quantitative “metric” to measure the level of code convergence has been implemented, with corresponding code diagnostics carried over from SOLPS4.3, allowing code run times to be accurately compared. Good agreement is found between the two codes in the tests so far. Remaining discrepancies in the solution in front of the divertor targets are likely related to different formulations of the equations and the different discretization schemes used by the codes.

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1. Introduction

The SOLPS plasma boundary code package is a combination of a 2D multi-fluid plasma transport code, B2 or B2.5, and the 3D kinetic Monte Carlo neutral transport code EIRENE [1]. In particular, SOLPS4.3 has been the workhorse for the ITER divertor design [2] and features the most sophisticated EIRENE version including neutral viscosity and line radiation opacity effects [3]. However, SOLPS4.3 uses B2 and thus cannot benefit from many refinements made to B2.5 by the fusion community over the past decade (e.g. new fluid drift formulation and a reworked 2D current equation). In order to unite the most recent advances in both the fluid and neutral codes, the ITER Organization (IO) has sponsored the development of a new package, named SOLPS-ITER, which couples the MPI parallelized EIRENE code with the B2.5 fluid plasma solver from the SOLPS5.2 package [4]. This new version is to be hosted by the IO and is to become the principal plasma boundary simulation tool of ITER.

Due to requirements for continuity in design activity, a critical issue is the backwards compatibility of SOLPS-ITER with previous ITER reference simulations performed using SOLPS4.3. Only after

a common reference point is established can additional physics effects and model refinements, which are available with the more general new code package, be isolated, publicly exposed and systematically investigated.

In this paper some of the challenges faced during the development of the new SOLPS-ITER code suite are highlighted. In the following we present a short summary of the dismantling procedure undertaken in particular for the benchmark with SOLPS4.3, in which the fluid and the Monte-Carlo parts of the codes were separately compared. In the main Section 5 we discuss the results of SOLPS-ITER simulations which have been performed on a selected series of ITER reference multi-fluid density-scans. The details of the comparison between SOLPS-ITER and SOLPS4.3 solutions are presented. We conclude the paper by discussing the ongoing verification results.

2. Analysis of the code differences

The implementations of the 2D plasma models in B2 and B2.5 have been previously analysed in detail in [5] for a single-fluid slab model application. This benchmark has now been extended to multi-fluid calculations, and the parameter settings required to reproduce the SOLPS4.3 simulations have been identified. Contrary to B2, B2.5 solves for internal electron and ion energy equations where the latter can be expressed by the following form (indeed,

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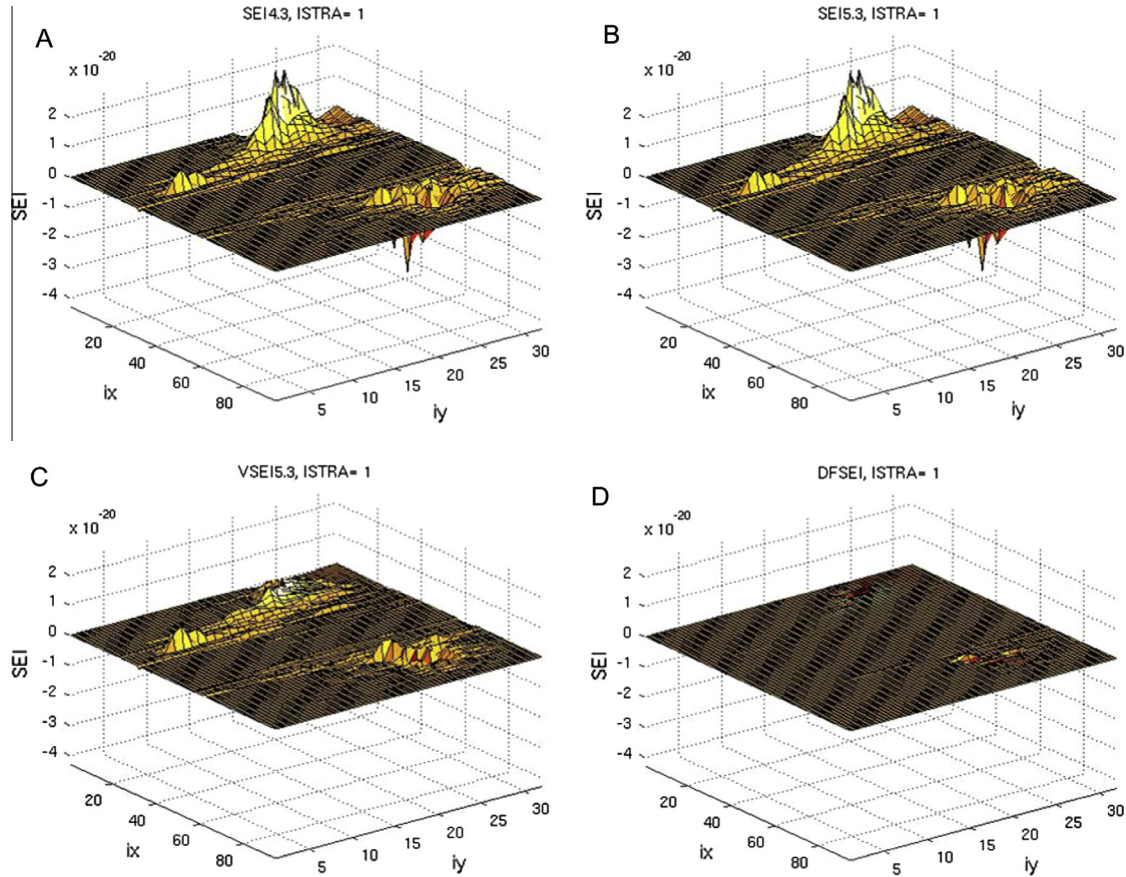


Fig. 1. Comparison of ion energy source terms calculated by EIRENE in SOLPS4.3 (A) and SOLPS-ITER (B) for a JET D + C case. The differences (D) between A and B are significantly smaller compared to the statistical error (C).

the analytical derivation in [5] can be easily extended also for multi-fluid systems):

$$\frac{\partial}{\partial t} \left(\frac{3}{2} n_i T_i \right) + \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} q_{ix} \right) + \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} q_{iy} \right) + \sum_{a=1}^{n_s} n_a T_i \frac{\partial}{\partial x} \left(\frac{\sqrt{g} b_x}{h_x} v_{a\parallel} \right) = S_{int}^i \quad (1)$$

which includes a sum over the divergences of parallel velocities for all species n_s . The $h_{x,y}$ and g define the metric coefficients of the curvilinear coordinate system, and S_{int}^i is the total internal energy source. The set of continuity, momentum and total energy equations is entirely equivalent to the set of continuity, momentum and internal energy equations if the source term due to neutrals from EIRENE, S_{eir} is corrected as follows:

$$S_{int}^i = S_{eir}^i + \sum_{a=1}^{n_s} \left[\frac{m_i}{2} v_{a\parallel}^2 S_a^n - v_{a\parallel} S_a^m \right] \quad (2)$$

where S_a^n and S_a^m are the particle and parallel momentum sources for species a , respectively. For the comparison with B2 we assume no drift and Pfirsch–Schlüter terms in the plasma flows and the zero current assumption was applied in both B2.5 and B2. Thus the heat fluxes in the poloidal and radial directions, respectively, are expressed as:

$$q_{ix} = \frac{3}{2} T_i \sum_{a=1}^{n_s} \Gamma_{ax} - \kappa_{ix} \frac{1}{h_x} \frac{\partial T_i}{\partial x}; \quad q_{iy} = \frac{3}{2} T_i \sum_{a=1}^{n_s} \Gamma_{ay} - \kappa_{iy} \frac{1}{h_y} \frac{\partial T_i}{\partial y} \quad (3)$$

Specification of appropriate code input parameters ensures that identical terms are included in the continuity and parallel momentum balance equations in B2 and B2.5. The latter discretizes the

parallel momentum equations on a collocated grid (parallel flow specified at cell centres), whereas B2 uses a staggered grid formulation (flow at cell boundaries).

For the benchmarking exercise described here, transport coefficients and other transport relevant parameters in B2.5 were adjusted to be the same as in B2, i.e. flux limited Braginskii classical parallel transport and radially constant anomalous transport coefficients. The boundary conditions for continuity and parallel momentum equations are identical to those in SOLPS4.3, and an equivalent formulation is used for the energy (i.e. by removing the kinetic part in the sheath heat transmission coefficients when specifying boundary conditions for the internal energy equations).

The B2.5 code has a model for fluid neutrals in place and it is important to ensure that there is no interference with the EIRENE kinetic neutral source terms, and that the ion energy equation is not affected when scaling the fluid neutral densities to very small values.

The reaction rates for other source terms in B2.5: impurity line-radiation, ionisation and recombination of non-hydrogenic ions, have been pre-processed using high-resolution data tables to ensure small differences in the percent range for the rate coefficients which, in B2, are derived using the analytical formulae of the STRAHL code.

In SOLPS-ITER, B2.5 has been coupled to the up-to-date version of MPI parallelized EIRENE. In the process of reviewing this coupling, care was taken to ensure that the EIRENE input specifications (neutral transport model and triangular geometry option) were fully backwards compatible with SOLPS4.3. We have ported the two-phased source rescaling scheme for the coupling as adopted in B2-EIRENE [2], allowing for low errors in global particle flux balances (critical for code convergence).

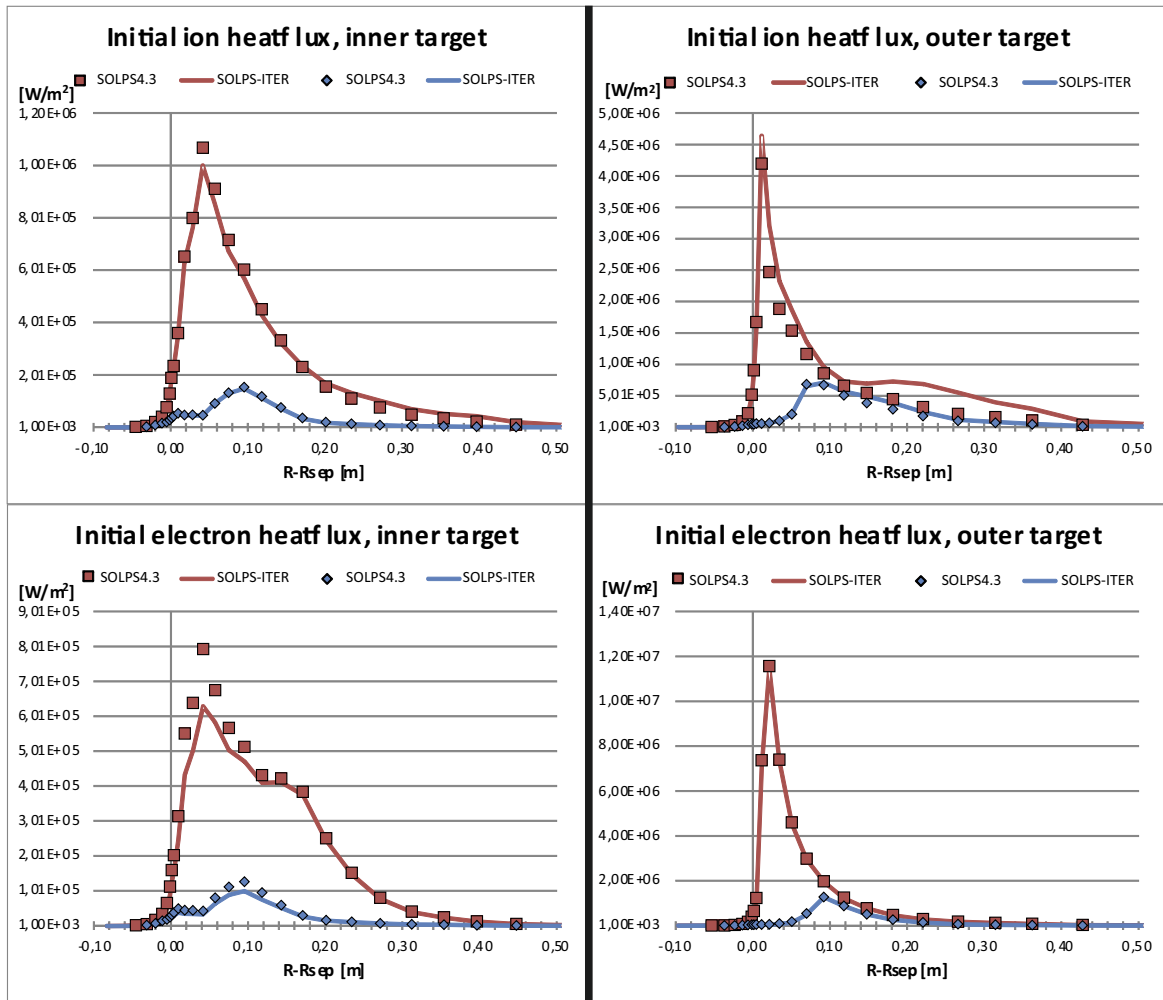


Fig. 2. Comparing target ion and electron heat flux discretizations from B2 with their reconstructions from B2.5, i.e. $q_i = q_i^{B2.5} + (T_i + E_{kin})\Gamma$ and $q_e = q_e^{B2.5} + T_e\Gamma$, respectively (neglecting a small viscous contribution to q_i), for low (red) and high (blue) throughput for $P_{edge} = 100$ MW. On fixed (initial) plasma backgrounds, the B2 fluxes (symbols) are approximately reproduced by B2.5. The discrepancies observed in the peak heat fluxes specifically for the low throughput (at higher target particle fluxes) can be related to differences in the implementations in B2 and B2.5 for the convective heat flow component in the hybrid discretization method [7], see text. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Basic benchmarks and tests

The previous section summarizes only very briefly the actions necessary to make SOLPS-ITER behave as similarly as possible in comparison with SOLPS4.3 and a more complete summary will be given in a separate paper. Likewise we highlight here at this point only those tests which turned out to be most critical ones for the verification of the new B2.5-EIRENE coupling in SOLPS-ITER.

We have employed a tool for converting a SOLPS4.3 plasma solution into SOLPS-ITER format [9] and passed this fixed plasma background to EIRENE via B2.5. The former has been shown to be backwards compatible down to the microscopic level of individual Monte Carlo particle trajectories, by employing “correlation sampling strategies” [6]. Furthermore an extended set of statistical diagnostics has been applied, including the transformation of neutral particle sources from the laboratory frame (B2) into the plasma frame (B2.5). Fig. 1 demonstrates that for a JET case the ion energy sources S_{eir}^i calculated by EIRENE in both SOLPS4.3 and SOLPS-ITER are indeed the same within statistical errors.

On the B2.5 side we implemented the same discretization schemes for the fluxes as those used in B2. For the particle fluxes, the modified upwind scheme is applied (Eq. (12) of [11]). For the same initial plasma density and velocities, B2.5 yields exactly the same particle fluxes as B2. However, for the electron and ion heat

fluxes, the discretizations are found to be different. By taking identical initial plasma parameters, Fig. 2 demonstrates that for a standard ITER case (F57 grid, cf. [13]) the discrepancy in the target heat fluxes between B2 and B2.5 cannot be eliminated, despite using the same hybrid discretization scheme (Eq. (11) of [11,7]). The ratio of convective to conductive heat flux, i.e. the Peclet number, enters the hybrid discretization scheme as a (local) parameter. The difference occurs because in the calculation for the convective heat flux, B2 uses a factor 5/2 and B2.5 a factor 3/2 (cf. Eq. (3)). We conclude that for the given ITER F57 grid used in these simulations, the discretization error for the heat fluxes does change when moving from the total (B2) to the internal (B2.5) energy equation and we note that the absolute discretization error using the hybrid method becomes smaller when Peclet number decreases with decreasing particle flux. Finally, we can only briefly mention here that for fully coupled B2.5/B2-EIRENE simulations a “code race” was conducted and similar code run times are observed assuming the same convergence criteria.

4. Code convergence

The matter of code convergence must not be underestimated and it is a common mistake to apply only lax concepts of convergence criteria. For credible ITER design modeling studies it was

stressed in [2] (and references therein) that suitable convergence criteria must be defined and adhered to under all circumstances. Otherwise the relevant operational parameters (for example the plasma particle throughput from source to sink, i.e. the pumped flux in the divertor) cannot be plausibly derived.

To allow an accurate comparison of the code run times and to measure the level of code convergence, a quantitative “metric” has been implemented in SOLPS-ITER with corresponding code diagnostics carried over from SOLPS4.3. We define the code to be converged if all of the following criteria are simultaneously met:

- General steadiness of: (a) plasma density and temperature at the separatrix; (b) effective charge (Z_{eff}); (c) total recombination rate.
- To measure the steadiness of global quantities the corresponding characteristic time-scales should be larger than a given lower limit, specifically for: (a) particle content > 0.5 s; (b) energy content > 3.0 s; (c) helium concentration at separatrix > 0.5 s.
- The global error in the net particle flux balance must not exceed 1%. Similarly, the error in the global power balance should be of order 1% only.

All of the above convergence criteria are critical. One more criterium is to be added when internal iterations on the fluid equations are executed, then the relative variation of the source rescaling factors per iteration should not exceed 10^{-4} .

5. Results of full benchmark

SOLPS-ITER has been set up to mimic two sets of existing SOLPS4.3 solutions for ITER: a combined particle throughput and power-scan ($P_{edge} = 60$ and 100 MW entering the plasma edge, coincident with the core boundary of the simulation grid) for a D + He + C ITER F57 geometry reference, assuming a narrow SOL width [12]. To speed up convergence times, a source term averaging scheme has been implemented in SOLPS-ITER as proposed in [8] to allow for a larger time-step ($\sim 10^{-5}$ s) to drive the simulation quickly towards the expected solution, followed by a final phase with a low time-step ($\sim 10^{-7}$ s), using internal iterations to further reduce the residuals. Through this process, the error in the global particle balances for all species is reduced to below few percent [2] and all simulations could be converged.

Fig. 3 compares the integrated electron and ion heat fluxes arriving at the targets as a function of particle throughput Γ_D and P_{edge} (note that only the plasma component is compared, the total incident heat flux also includes the energy of ion recombination, potential and kinetic energy of incident neutral particles and radiation). We observe that SOLPS-ITER reproduces the Γ_D dependency in general, and that the two codes are in good agreement at

high Γ_D . However, at low Γ_D we observe significantly larger heat fluxes at the outer target in SOLPS-ITER. This discrepancy is larger for higher P_{edge} and stronger for the ion heat channel.

The code is generally able to reproduce the upstream conditions for density and temperature (Fig. 4) and although we observe a slight deviation from the reference density profile in the far-SOL wings at low Γ_D , the particle fluxes at both targets are reasonably well matched. For this case SOLPS-ITER does not reproduce the peaked structure in the electron temperature profile close to the outer target separatrix at high P_{edge} and low Γ_D . In common with the integrated heat fluxes, we observe a discrepancy in the peak heat fluxes of up to a factor 2 at the outer target for low Γ_D .

The discrepancies become generally smaller with increasing Γ_D (or decrease in P_{edge}). At the same P_{edge} but high Γ_D , the differences in the target particle fluxes between the two codes appear more prominent, leading to a stronger peaking in the density for SOLPS-ITER. However, and in contrast to the low Γ_D case, the target heat fluxes are now in much better agreement. Consequently, target temperature profiles coincide in general at high Γ_D , although the absolute values of ion temperature and heat flux are still slightly higher.

6. Discussion and conclusions

In the ITER simulations the upstream plasma parameters calculated with SOLPS4.3 and SOLPS-ITER are found to be in good agreement. Only by imposing the same convergence “metric” as in SOLPS4.3 is the same approximately true also for the divertor. However at the divertor targets, quantitative discrepancies can reach a factor 2 in the extreme case (low throughput, high power). With increasing particle throughput or decreasing power the disagreements become smaller (and are comparable to the results previously found in the single-fluid low-power JET benchmarks when comparing SOLPS4.3 and SOLPS5.0 [10]).

We have made a major effort to demonstrate that the EIRENE component in SOLPS-ITER produces identical results to SOLPS4.3 for the source terms coupled into B2.5. We have also been able to show that the B2.5 plasma fluid transport model can be made identical to that in B2 by extending the previous effort in [5] towards multi-fluid simulations, taking into account the equivalence of the total energy equations (B2) with the internal energy (B2.5) equations.

The remaining differences between the solutions found in the divertor can be related to different discretization schemes; discretization errors and different implementation of curvilinear coordinates and non-orthogonal grids between the two code packages (the in-depth B2–B2.5 code disintegration exercise has only to date been performed a slab model [5]). As a consequence, we suggest as a next step to extend the SOLPS-ITER benchmark for ITER simulations

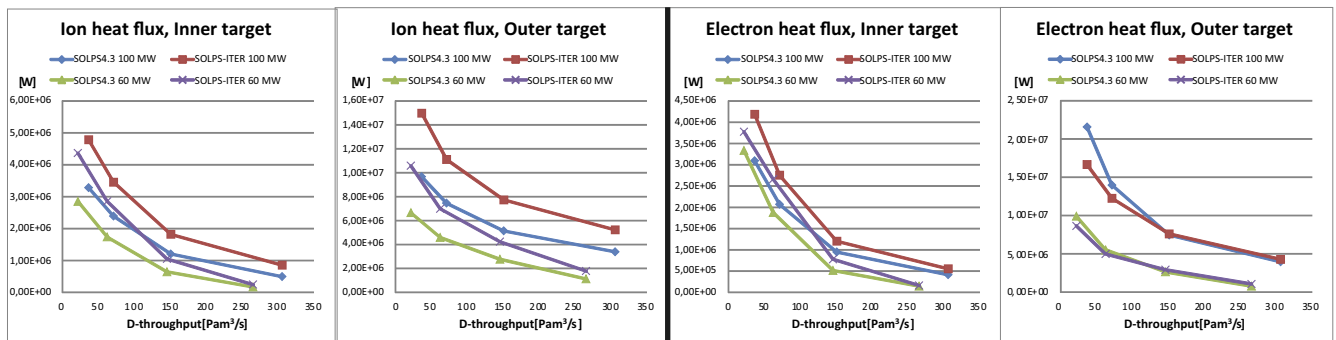


Fig. 3. Integrated target ion and electron heat fluxes for the F57 D + He + C ITER power and throughput scan (square symbols and diamonds: 100 MW, triangles and crosses: 60 MW). In absolute numbers the differences between SOLPS4.3 and SOLPS-ITER are larger at the outer target and become smaller towards higher throughput or lower power.

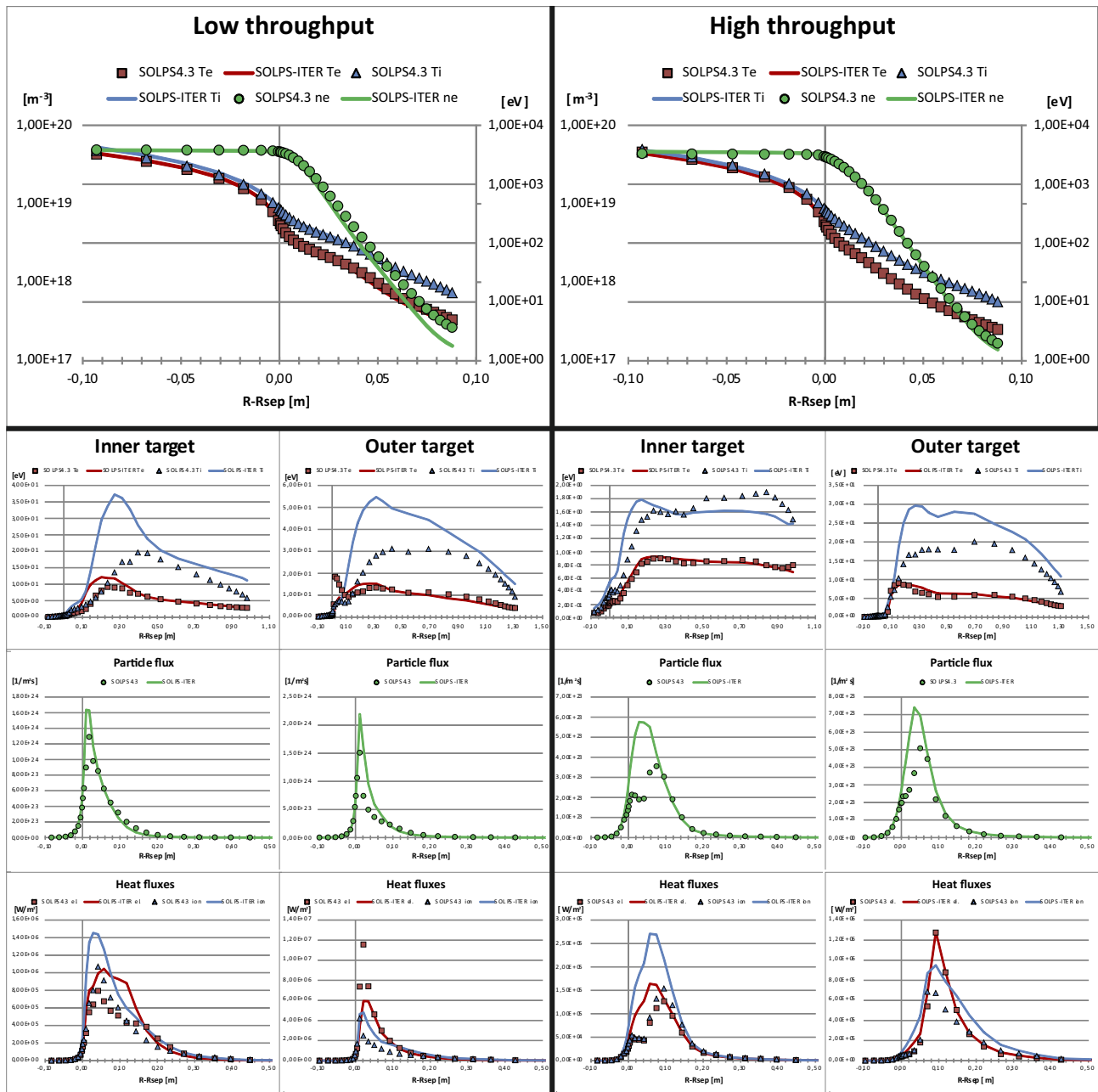


Fig. 4. Detailed comparison of the 100 MW F57 ITER D + He + C case. From top: outer mid-plane temperature and density profiles, target temperature profiles, particle fluxes and electron and ion heat fluxes, at low and high particle throughput (note that the target fluxes are displayed with a shorter abscissa compared to the target temperature plots).

by executing a detailed discretization error analysis, i.e. to compare the results for ITER multi-fluid SOLPS4.3 simulations with SOLPS-ITER using higher grid resolutions on both sides. It is finally suggested that the convergence criteria “metric” should be extended by a given limit for the flux discretization error.

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